Gingerbread Float Software

November 2026, Paul A. Cornelius

Platform

The program runs on a dedicated Raspberry PI 4. The light strings are controlled by two Falcon PixelController V5s, which are connected to the PI through a network hub. This hardware is physically located inside the float's "house." An Ethernet connector on the right side of the house can be used to access the float's network from an external computer.

Raspberry PI setup

The details are in ginger/rpi/FOL config. This file includes the IP addresses for the PI and the PixelControllers. There is no DMS service on this simple LAN, so all three units have fixed IP addresses. The PixelController addresses are hard-coded in the program so they must not be changed. They are set initially through the PixelController's on-board interface.

When power is switched on to the PI it runs the program automatically (as soon as the OS finishes initializing).

PixelController setup

The two PixelControllers need to be configured exactly or else the program will not work properly. The software and the hardware are closely linked. The PixelController configurations are saved in two json files in python3.14/ginger.

PixelController #1 (@192.168.1.20)

Controls 14 light strings.

- The River is 6 strings of 250 lights each, 5 volts.
- The 4 Candy Canes on the front of the float each have 2 strings of 60 lights, 12 volts.

PixelController #2 (@192.168.1.21)

- The 4 Candy Canes near the back of the float each have 1 string of 60 lights, 12 volts.
- The eves of the house have 2 strings of 100 lights each, 12 volts.

Software

The software lives on the Raspberry PI in "/home/fol/pyroot/ginger/main.py". It imports a utility package named PACN. A brief functional description follows. See the documentation for the PACN library for definitions of the terms used here.

River

The design concept is a “candy river.” It flows across the float from right to left. The lights therefore show a somewhat wavy bluish background, with random waves of color entering from the right. The background pattern moves more slowly than the color bands. Most of the river parameters are adjustable, and not all strings travel at the same rate. The parameters in the program were the result of some experimentation and were generally deemed acceptable.

Each string of river lights is 250 pixels long. Since PACN universes are limited to 170 pixels, each string requires 2 universes to hold its state.

Candy Canes (front)

These are just a simple red-and-white pattern that travels up the cane. All of the canes follow the same strategy. The only complication is that each cane is two strings laid out back-to-front. The software moves the pattern from one cane to another, treating the two 60-pixel strings as a single 120-pixel object.

Candy Canes (rear)

This is the same routine as the front canes, except that these canes are single 60-pixel strings.

Eve lights

These strings are not dynamic. They just come on to full white.

Software github repositories